

CALCULUS II

WEEK VI

THE DOT PRODUCT AND PROJECTIONS

→ Given two vectors $\vec{u} = u_1\vec{i} + u_2\vec{j}$ and $\vec{v} = v_1\vec{i} + v_2\vec{j}$ in $\mathbb{R}^2$, we define their dot product $u \cdot v$ to be the sum of the products of their corresponding components.

$$\boxed{\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2}$$

→ The terms scalar product and inner product are also used in place of dot product. Similarly, for vectors $\vec{u} = u_1\vec{i} + u_2\vec{j} + u_3\vec{k}$ and $\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$ in $\mathbb{R}^3$.

$$\boxed{\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2 + u_3v_3}$$

Properties of the Dot Product

→ The dot product has the following algebraic properties

$$\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u} \quad (\text{commutative law})$$

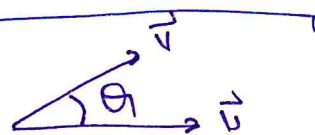
$$\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w} \quad (\text{distributive law})$$

$$(t\vec{u}) \cdot \vec{v} = \vec{u} \cdot (t\vec{v}) = t(\vec{u} \cdot \vec{v}) \quad t \in \mathbb{R} \quad (\text{for real } t)$$

$$\vec{u} \cdot \vec{u} = |\vec{u}|^2$$

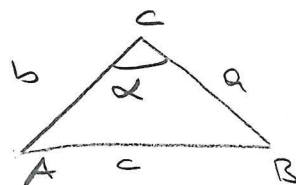
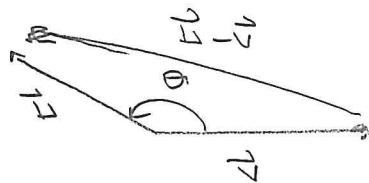
Theorem If θ is the angle between the directions of $\vec{u}$ and $\vec{v}$ ($0 \leq \theta \leq \pi$), then

$$\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$$



In particular, $\vec{u} \cdot \vec{v} = 0$ if and only if $\vec{u}$ and $\vec{v}$ are perpendicular. $\vec{u} \perp \vec{v}$ iff $\vec{u} \cdot \vec{v} = 0$.
(Of course, the zero vector is perpendicular to every vector.)

Proof Applying the cosine law to a triangle reveals the relationship between dot the product and angle between vectors



cosine law

$$c^2 = a^2 + b^2 - 2ab \cos \alpha$$

$$\begin{aligned} |\vec{u} - \vec{v}|^2 &= |\vec{u}|^2 + |\vec{v}|^2 - 2|\vec{u}||\vec{v}| \cos \theta \\ &= (\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v}) \\ &= \vec{u} \cdot \vec{u} - \vec{u} \cdot \vec{v} - \vec{v} \cdot \vec{u} + \vec{v} \cdot \vec{v} \\ &= |\vec{u}|^2 - 2\vec{u} \cdot \vec{v} + |\vec{v}|^2 \end{aligned}$$

$$\begin{aligned} |\vec{u}|^2 + |\vec{v}|^2 - 2|\vec{u}||\vec{v}| \cos \theta &= \\ |\vec{u}|^2 + |\vec{v}|^2 - 2\vec{u} \cdot \vec{v} & \end{aligned}$$

$$\vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}| \cos \theta$$

$$\cos \theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}||\vec{v}|}$$

Hence, $\vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}| \cos \theta$

Ex Find the angle θ between the vectors $\vec{u} = 2\vec{i} + \vec{j} - 2\vec{k}$ and $\vec{v} = 3\vec{i} - 2\vec{j} - \vec{k}$

Solving the formula $\vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}| \cos \theta$ for θ , we obtain

$$\theta = \arccos \left(\frac{\vec{u} \cdot \vec{v}}{|\vec{u}||\vec{v}|} \right) = \arccos \left(\frac{2 \cdot 3 + 1 \cdot (-2) + (-2) \cdot (-1)}{3\sqrt{14}} \right)$$

$$= \arccos \left(\frac{2}{\sqrt{14}} \right) \approx 57.69^\circ$$

Parallel vectors If two nonzero vectors $\vec{u}$ and $\vec{v}$ are parallel if and only if $\vec{u} = c\vec{v}$ for $c \in \mathbb{R}$. If $\theta = 0$ $c > 0$, $\theta = \pi$ $c < 0$

- Ex
- (a) $\langle 2, 4 \rangle \cdot \langle -1, 5 \rangle = -2 + 20 = 18$
 - (b) $\langle -1, 7, 4 \rangle \cdot \langle 6, 2, -1/2 \rangle = -6 + 14 - 2 = 6$
 - (c) $(\vec{i} + 2\vec{j} - 4\vec{k}) \cdot (2\vec{j} - \vec{k}) = 1 \cdot 0 + 2 \cdot 2 - 4 \cdot (-1) = 6$

Ex Determine if the following vectors are parallel, orthogonal or neither

(a) $\vec{a} = \langle 6, -2, -1 \rangle$ $\vec{b} = \langle 2, 5, 2 \rangle$

$$\vec{a} \cdot \vec{b} = 6 \cdot 2 - 2 \cdot 5 - 1 \cdot 2 = 0$$

$$\vec{a} \perp \vec{b}$$

(b) $\vec{u} = 2\vec{i} - \vec{j}$ $\vec{v} = -\frac{1}{2}\vec{i} + \frac{1}{4}\vec{j}$

$$\vec{u} = -4\vec{v} \quad \vec{u} \text{ and } \vec{v} \text{ are parallel}$$

or

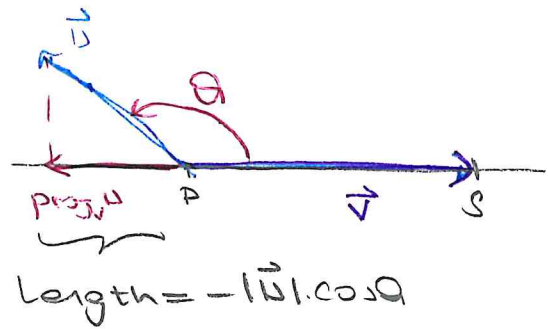
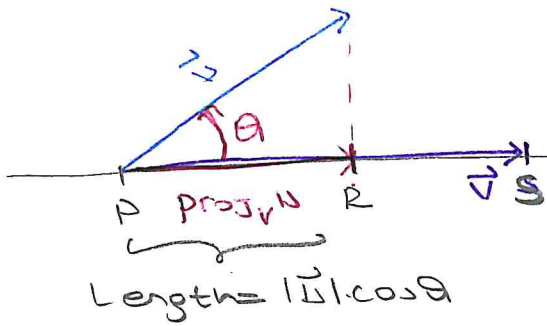
Since $\vec{u} \cdot \vec{v} = 2(-\frac{1}{2}) - 1(\frac{1}{4}) = -5/4$ $|\vec{u}| = \sqrt{5}$ $|\vec{v}| = \sqrt{5}/4$ so

$$-\frac{5}{4} = \sqrt{5} \frac{\sqrt{5}}{4} \cos \theta$$

$$\cos \theta = -1 \quad \theta = \pi$$

two vectors are parallel.

Projections



$$\text{proj}_{\vec{V}} \vec{U} = \underbrace{(|\vec{U}| \cos \theta)}_{\frac{\vec{U} \cdot \vec{V}}{|\vec{V}|}} \frac{\vec{V}}{|\vec{V}|} = \left(\frac{\vec{U} \cdot \vec{V}}{|\vec{V}|^2} \right) \vec{V} = \frac{\vec{U} \cdot \vec{V}}{|\vec{V}|^2} \vec{V}$$

$$\left(|\vec{U}| \cos \theta = \frac{|\vec{U}| \cdot |\vec{V}| \cos \theta}{|\vec{V}|} = \frac{\vec{U} \cdot \vec{V}}{|\vec{V}|} \right)$$

The vector projection of $\vec{U}$ onto $\vec{V}$ is the vector

$$\text{proj}_{\vec{V}} \vec{U} = \left(\frac{\vec{U} \cdot \vec{V}}{|\vec{V}|^2} \right) \vec{V}$$

The scalar component of $\vec{U}$ in the direction of $\vec{V}$ is the scalar

$$\text{Scalar}_{\vec{V}} \vec{U} = |\vec{U}| \cos \theta = \frac{\vec{U} \cdot \vec{V}}{|\vec{V}|} = \vec{U} \cdot \frac{\vec{V}}{|\vec{V}|}$$

$$\text{proj}_{\vec{V}} \vec{U} = \text{Scalar}_{\vec{V}} \vec{U} \cdot \frac{\vec{V}}{|\vec{V}|}$$

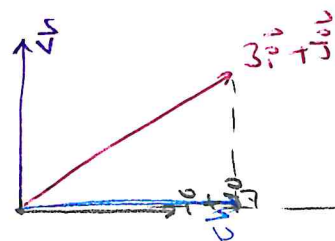
Ex 7 Express the vector $3\vec{i} + \vec{j}$ as a sum of vectors $\vec{U} + \vec{V}$ where $\vec{U}$ is parallel to the vector $\vec{i} + \vec{j}$ and $\vec{V}$ is perpendicular to $\vec{U}$.

METHOD I (Using vector projection) Note that $\vec{U}$ must be the vector projection of $3\vec{i} + \vec{j}$ in the direction of $\vec{i} + \vec{j}$. Thus,

$$\vec{U} = \frac{(3\vec{i} + \vec{j}) \cdot (\vec{i} + \vec{j})}{(\vec{i} + \vec{j}) \cdot (\vec{i} + \vec{j})} (\vec{i} + \vec{j}) = \frac{4}{2} (\vec{i} + \vec{j}) = 2\vec{i} + 2\vec{j}$$

$$\vec{U} + \vec{V} = 3\vec{i} + \vec{j} \quad |\vec{i} + \vec{j}|^2$$

$$\vec{V} = 3\vec{i} + \vec{j} - \vec{U} = 3\vec{i} + \vec{j} - (2\vec{i} + 2\vec{j}) = \vec{i} - \vec{j}$$



$$\vec{V} = \vec{a} - \vec{U} = \vec{a} - \frac{(\vec{a} \cdot \vec{b})}{|\vec{b}|^2} \vec{b}$$

$$\vec{V} \cdot \vec{b} = \vec{a} \cdot \vec{b} - \frac{(\vec{a} \cdot \vec{b})}{|\vec{b}|^2} (\vec{b} \cdot \vec{b}) = \vec{a} \cdot \vec{b} - \frac{(\vec{a} \cdot \vec{b})^2}{|\vec{b}|^2} = 0$$

METHOD II Since $\vec{u}$ is parallel to $\vec{i} + \vec{j}$ and $\vec{v}$ is perpendicular

to $\vec{u}$, we have

$$\vec{u} = t(\vec{i} + \vec{j}) \quad \text{and} \quad \vec{v} \cdot (\vec{i} + \vec{j}) = 0$$

for some scalar t . We want $\vec{u} + \vec{v} = 3\vec{i} + \vec{j}$. Take the dot product of this equation with $\vec{i} + \vec{j}$

$$\vec{u} \cdot (\vec{i} + \vec{j}) + \underbrace{\vec{v} \cdot (\vec{i} + \vec{j})}_0 = (3\vec{i} + \vec{j}) \cdot (\vec{i} + \vec{j})$$

$$t(\vec{i} + \vec{j}) \cdot (\vec{i} + \vec{j}) + 0 = 4$$

$$2t = 4, \quad t = 2. \quad \text{Therefore, } \vec{u} = 2\vec{i} + 2\vec{j} \quad \text{and} \quad \vec{v} = 3\vec{i} + \vec{j} - \vec{u} = \vec{i} - \vec{j}$$

Slope of a vector Let $\vec{v} = a\vec{i} + b\vec{j}$. then the slope of $\vec{v} = \frac{b}{a} = \tan \theta$

- A vector is tangent or normal to a curve at a point if it is parallel or normal to the line that is tangent to the curve at that point.

Ex Find unit vectors tangent and normal to the curve

$$y = \frac{x^3}{2} + \frac{1}{2} \quad \text{at the point } (1, 1)$$

$$y = \frac{x^3}{2} + \frac{1}{2} \quad \frac{dy}{dx} \bigg|_{(1,1)} = \frac{3x^2}{2} \bigg|_{(1,1)} = \frac{3}{2} \quad \text{the slope of the tangent line to the curve at } (1,1)$$

Unit tangent vector we will find a unit vector with this slope.

$$\vec{v} = a\vec{i} + b\vec{j} = 2\vec{i} + 3\vec{j} \quad \text{and} \quad \text{every nonzero multiple of } \vec{v} \text{ has}$$

$$\text{Slope} \quad \frac{3}{2} = \frac{b}{a}. \quad \text{But this vector is not a unit vector.}$$

Divide it by $|\vec{v}|$ to get the unit tangent vector.

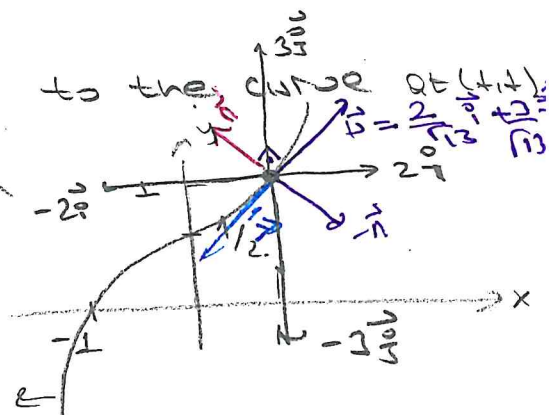
$$\vec{u} = \frac{\vec{v}}{|\vec{v}|} = \frac{2\vec{i} + 3\vec{j}}{\sqrt{4+9}} = \frac{2}{\sqrt{13}}\vec{i} + \frac{3}{\sqrt{13}}\vec{j}$$

Remark $-\vec{u} = -\frac{2}{\sqrt{13}}\vec{i} - \frac{3}{\sqrt{13}}\vec{j}$ is also tangent to the curve at $(1,1)$.

which points in the opposite direction

(4)

$$y = \frac{x^3}{2} + \frac{1}{2}$$



Unit normal vector To find unit vectors normal to the curve at $(1,1)$ we look for unit vectors whose slope is $-\frac{2}{3}$. Hence

$$\vec{n} = -\frac{3}{\sqrt{13}}\vec{i} + \frac{2}{\sqrt{13}}\vec{j} \quad (\text{or} \quad -\vec{n} = \frac{3}{\sqrt{13}}\vec{i} - \frac{2}{\sqrt{13}}\vec{j})$$

$\vec{D} \cdot \vec{n} = 0$ (they are perpendicular to each other)

VECTORS IN n-SPACE

Vectors in $\mathbb{R}^n$ can be expressed as linear combinations of the n unit vectors

$\vec{e}_1$ from the origin to the point $(1, 0, 0, \dots, 0)$
 $\vec{e}_2$ from the origin to the point $(0, 1, 0, \dots, 0)$
 $\vec{e}_n$ from the origin to the point $(0, 0, 0, \dots, 1)$

These vectors constitute a standard basis in $\mathbb{R}^n$. The n -vector $\vec{x}$ with components $x_1, x_2, \dots, x_n$ is expressed in the form

→ The $\vec{x} = (x_1, x_2, \dots, x_n)$ is expressed in the form

$$\vec{x} = x_1\vec{e}_1 + x_2\vec{e}_2 + \dots + x_n\vec{e}_n$$

The length of $\vec{x}$ is $|\vec{x}| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$

→ The angle between $\vec{x} = (x_1, x_2, \dots, x_n)$ and $\vec{y} = (y_1, y_2, \dots, y_n)$

is

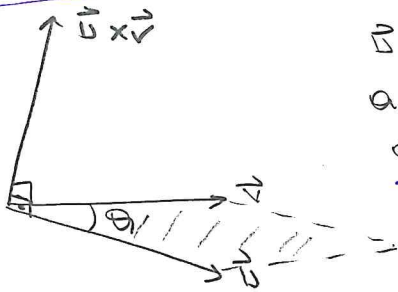
$$\theta = \arccos \left(\frac{\vec{x} \cdot \vec{y}}{|\vec{x}| |\vec{y}|} \right)$$

where $\vec{x} \cdot \vec{y} = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$

THE CROSS PRODUCT IN 3-SPACE

For any vectors $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^3$, the cross-product $\vec{u} \times \vec{v}$ is the unique vector satisfying the following three conditions

- (i) $(\vec{u} \times \vec{v}) \cdot \vec{u} = 0$ and $(\vec{u} \times \vec{v}) \cdot \vec{v} = 0$
- (ii) $|\vec{u} \times \vec{v}| = |\vec{u}| |\vec{v}| \sin \theta$ where θ is the angle between $\vec{u}$ and $\vec{v}$
- (iii) $\vec{u}, \vec{v}$ and $\vec{u} \times \vec{v}$ form a right-handed triad



$\vec{u} \times \vec{v}$ is perpendicular to both $\vec{u}$ and $\vec{v}$ and has length equal to the area of the shaded parallelogram

Right hand rule (Find direction $\vec{u} \times \vec{v}$)

If the fingers of your right hand curl in the direction of a rotation (through angle less than π) from $\vec{u}$ to $\vec{v}$, then your thumb points the $\vec{u} \times \vec{v}$ direction.

Parallel Vectors

Nonzero

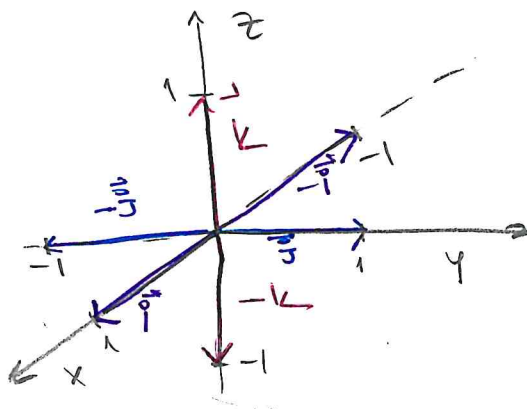
vectors $\vec{u}$ and $\vec{v}$ are parallel if and only if $\vec{u} \times \vec{v} = 0$

$$\vec{u} \parallel \vec{v} \Leftrightarrow \vec{u} \times \vec{v} = 0$$

Properties of the Cross Product

If $\vec{u}, \vec{v}$ and $\vec{w}$ are any vectors and r, s are scalars, then

- (1) $(r\vec{u}) \times (s\vec{v}) = (rs)(\vec{u} \times \vec{v})$
- (2) $\vec{u} \times (\vec{v} + \vec{w}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{w}$
- (3) $(\vec{v} \times \vec{u}) = -(\vec{u} \times \vec{v})$
- (4) $(\vec{v} + \vec{w}) \times \vec{u} = (\vec{v} \times \vec{u}) + (\vec{w} \times \vec{u})$
- (5) $0 \times \vec{u} = 0$
- (6) $\vec{u} \times (\vec{v} \times \vec{w}) = (\underbrace{\vec{u} \cdot \vec{w}}_{\text{inner product}}) \vec{v} - (\underbrace{\vec{u} \cdot \vec{v}}_{\text{inner product}}) \vec{w}$



$\vec{i} \times \vec{i} = 0$	$\vec{i} \times \vec{j} = \vec{k}$	$\vec{j} \times \vec{i} = -\vec{k}$
$\vec{j} \times \vec{j} = 0$	$\vec{j} \times \vec{k} = \vec{i}$	$\vec{k} \times \vec{j} = -\vec{i}$
$\vec{k} \times \vec{k} = 0$	$\vec{k} \times \vec{i} = \vec{j}$	$\vec{i} \times \vec{k} = -\vec{j}$

Inner product

($\vec{i} \cdot \vec{i} = 1, \vec{j} \cdot \vec{j} = 1, \vec{k} \cdot \vec{k} = 1$)

($\vec{i} \cdot \vec{j} = 0, \vec{j} \cdot \vec{k} = 0, \vec{k} \cdot \vec{i} = 0$)

($\vec{i} \cdot \vec{k} = 0, \vec{j} \cdot \vec{i} = 0, \vec{k} \cdot \vec{j} = 0$)

($\vec{i} \cdot \vec{i} = 1, \vec{j} \cdot \vec{j} = 1, \vec{k} \cdot \vec{k} = 1$)

($\vec{i} \cdot \vec{j} = 0, \vec{j} \cdot \vec{k} = 0, \vec{k} \cdot \vec{i} = 0$)

($\vec{i} \cdot \vec{k} = 0, \vec{j} \cdot \vec{i} = 0, \vec{k} \cdot \vec{j} = 0$)

$$(2\vec{i} + \frac{5}{3}\vec{j} - 3\vec{k}) \times (-2\frac{5}{3}\vec{j} + 5\vec{k})$$

$$= ((1)(5) - (-2)(-3))\vec{i} + ((-3)(0) - (2)(5))\frac{5}{3}\vec{j} + ((2)(-2) - (1)(0))\vec{k}$$

$$= -\vec{i} - 10\frac{5}{3}\vec{j} - 4\vec{k}$$

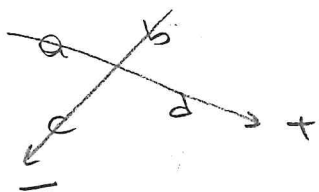
THM Components of the cross product

If $\vec{U} = u_1\vec{i} + u_2\vec{j} + u_3\vec{k}$ and $\vec{V} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$, then

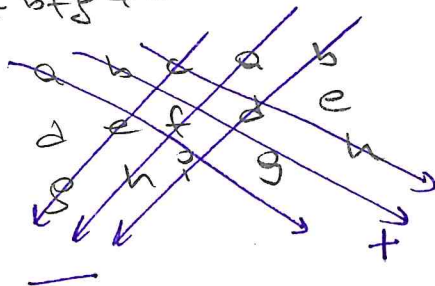
$$\vec{U} \times \vec{V} = (u_2v_3 - u_3v_2)\vec{i} + (u_3v_1 - u_1v_3)\vec{j} + (u_1v_2 - u_2v_1)\vec{k}$$

Determinants

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$



$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - gec - hfa - idb$$



$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

$$= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

Minors

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + e \begin{vmatrix} a & c \\ g & i \end{vmatrix} - h \begin{vmatrix} a & c \\ d & f \end{vmatrix}$$

The pattern of + and - signs used in 3x3 matrices

$$\begin{vmatrix} + & - & + \\ - & + & - \end{vmatrix}$$

(7)

Ex $\begin{vmatrix} 1 & 4 & -2 \\ -3 & 1 & 0 \\ 2 & 2 & -3 \end{vmatrix} = 3 \begin{vmatrix} 4 & -2 \\ 2 & -3 \end{vmatrix} + 1 \begin{vmatrix} 1 & -2 \\ 2 & -3 \end{vmatrix}$
 $= 3(-4) + 1 = -23.$

Properties of Determinants

(P) If two rows of a determinant are interchanged, then the determinant changes sign

$$\begin{vmatrix} d & e & f \\ a & b & c \\ g & h & i \end{vmatrix} = - \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

(II) If two rows of a determinant are equal, the determinant has value 0

$$\begin{vmatrix} a & b & c \\ a & b & c \\ g & h & i \end{vmatrix} = 0$$

(III) If a multiple of one row of a determinant is added to another row, the value of the determinant remains unchanged

$$\begin{vmatrix} a & b & c \\ d+ta & e+tb & f+tc \\ g & h & i \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

The Cross Product as Determinant

$$\vec{U} = u_1 \vec{i} + u_2 \vec{j} + u_3 \vec{k}$$

$$\vec{V} = v_1 \vec{i} + v_2 \vec{j} + v_3 \vec{k}$$

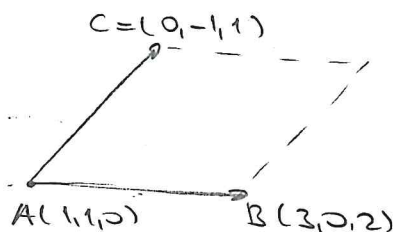
$$\vec{U} \times \vec{V} = (u_2 v_3 - u_3 v_2) \vec{i} + (u_3 v_1 - u_1 v_3) \vec{j} + (u_1 v_2 - u_2 v_1) \vec{k}$$

$$\vec{U} \times \vec{V} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \vec{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \vec{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \vec{k}$$

Ex 7 Find the area of the triangle with vertices at the three points $A = (1, 1, 0)$, $B = (3, 0, 2)$ and $C = (0, -1, 1)$

$$\vec{AB} = 2\vec{i} - \vec{j} + 2\vec{k}$$

$$\vec{AC} = -\vec{i} - 2\vec{j} + \vec{k}$$



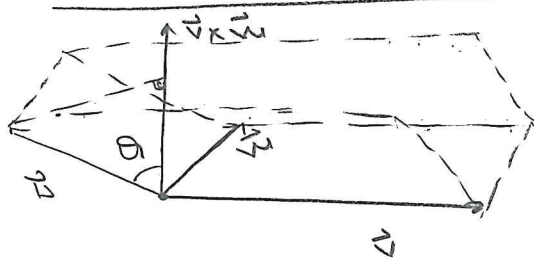
The area of the triangle is half the area of the parallelogram spanned by $\vec{AB}$ and $\vec{AC}$. By the definition of the cross product

$$\frac{1}{2} |\vec{AB} \times \vec{AC}| = \frac{1}{2} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 2 \\ -1 & -2 & 1 \end{vmatrix}$$

$$= \frac{1}{2} |3\vec{i} - 4\vec{j} - 5\vec{k}| = \frac{1}{2} \sqrt{16 + 9 + 25} = \frac{5\sqrt{2}}{2} \text{ square units}$$

The Cross Product as a Determinant

A parallelepiped is the three-dimensional analogue of a parallelogram. It is a solid with three pairs of parallel planar faces. Each face is in the shape of a parallelogram.



We say that a parallelepiped is spanned by three vectors coinciding with three of its edges that meet at one vertex (origin).

$$\text{Volume of parallelepiped} = |\vec{w}| |\vec{v} \times \vec{u}| |\cos \theta| = |\vec{w} \cdot (\vec{v} \times \vec{u})| \text{ cubic units}$$

The quantity $\vec{w} \cdot (\vec{v} \times \vec{u})$ is called the scalar triple product of the vectors $\vec{w}$, $\vec{v}$ and $\vec{u}$.

The Cross Product as a Determinant

→ The scalar triple product is easily expressed in terms of a determinant. If $\vec{u} = u_1\vec{i} + u_2\vec{j} + u_3\vec{k}$ and similar representations hold for $\vec{v}$ and $\vec{w}$, then

$$\begin{aligned}\vec{u} \cdot (\vec{v} \times \vec{w}) &= u_1 \begin{vmatrix} v_2 & v_3 \\ w_2 & w_3 \end{vmatrix} - u_2 \begin{vmatrix} v_1 & v_3 \\ w_1 & w_3 \end{vmatrix} + u_3 \begin{vmatrix} v_1 & v_2 \\ w_1 & w_2 \end{vmatrix} \\ &= \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}\end{aligned}$$

→ The volume of the parallelepiped spanned by $\vec{u}$, $\vec{v}$ and $\vec{w}$ is the absolute value of this determinant.

→ Using the properties of the determinant, it is easily verified that

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = \vec{v} \cdot (\vec{w} \times \vec{u}) = \vec{w} \cdot (\vec{u} \times \vec{v})$$

→ Three vectors in 3-space are said to be coplanar if the parallelepiped they span has zero volume. If their tails coincide, three such vectors must lie in the same plane.

$$\vec{u}, \vec{v} \text{ and } \vec{w} \text{ are coplanar} \Leftrightarrow \vec{u} \cdot (\vec{v} \times \vec{w}) = 0$$

→ If $\vec{u}, \vec{v}$ and $\vec{w}$ are in the same plane, volume = 0

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = 0$$

→ parallelepiped

$$\Leftrightarrow \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = 0$$

→ Three vectors are certainly coplanar if any of them is 0 or if any pair of them is parallel.

→ If neither of these degenerate conditions apply, they are only coplanar if any one of them can be expressed as a linear combination of the other two.

Thm The vector $\vec{a} \times \vec{b}$ is orthogonal to both $\vec{a}$ and $\vec{b}$

Proof Consider $(\vec{a} \times \vec{b}) \cdot \vec{a}$

$$\begin{aligned}(\vec{a} \times \vec{b}) \cdot \vec{a} &= a_1(a_2b_3 - a_3b_2) - a_2(a_1b_3 - a_3b_1) + a_3(a_1b_2 - a_2b_1) \\&= a_1a_2b_3 - a_1a_3b_2 - a_2a_1b_3 + a_2a_3b_1 + a_3a_1b_2 - a_3a_2b_1 \\&= 0\end{aligned}$$

Similarly, $(\vec{a} \times \vec{b}) \cdot \vec{b} = 0$

Hence, the vector $\vec{a} \times \vec{b}$ is orthogonal to both $\vec{a}$ and $\vec{b}$

Remark This theorem says that if $\vec{a}$ and $\vec{b}$ are represented by directed line segments with the same initial point, the cross product $\vec{a} \times \vec{b}$ points in a direction perpendicular to the plane through $\vec{a}$ and $\vec{b}$.

Thm If θ is the angle between $\vec{a}$ and $\vec{b}$ ($0 \leq \theta \leq \pi$)

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$$

Proof $|\vec{a} \times \vec{b}|^2 = (a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2$

$$\begin{aligned}&= a_2^2b_3^2 - 2a_2b_3a_3b_2 + a_3^2b_2^2 + a_3^2b_1^2 - 2a_3b_1a_1b_3 \\&\quad + a_1^2b_3^2 + a_1^2b_2^2 - 2a_1a_2b_1b_2 + a_2^2b_1^2 \\&= (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2 \\&= |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2 |\vec{b}|^2 - |\vec{a}|^2 |\vec{b}|^2 \cos^2 \theta \\&= |\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta\end{aligned}$$

$$(\sqrt{\sin^2 \theta} = \sin \theta \quad \theta \in [0, \pi])$$

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$$

Corollary Two nonzero vectors $\vec{a}$ and $\vec{b}$ are parallel if and

only if $\vec{a} \times \vec{b} = \vec{0}$

Two nonzero vectors $\vec{a}$ and $\vec{b}$ are parallel if and

only if $\theta = 0$ or $\theta = \pi$. In either case $\sin \theta = 0$

$$|\vec{a} \times \vec{b}| = 0 \Rightarrow \vec{a} \times \vec{b} = \vec{0}$$